

Alexander Bernstein

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Education

University of California, Santa Barbara

Ph.D. in Statistics

Santa Barbara, CA

Sept. 2025

Dissertation: “Long-Only Minimum Variance Portfolio Composition for Factor Models”

Washington University in St. Louis

M.Sc. in Systems Science and Applied Mathematics

St. Louis, MO

Dec. 2016

B.A. in Mathematics and Economics (Cum Laude)

May 2014

Technical Skills

Languages & Tools: Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, CVXPY), SQL, R, C, Java, JavaScript/NodeJS, FastAPI, Flask, REST APIs, Docker, Git, Linux, AWS, MongoDB, L^AT_EX.

Machine Learning: Neural Networks, Gaussian Processes, Reinforcement Learning (Q-Learning), Ensemble Methods (Boosting/Bagging), Bayesian Inference, Time Series Forecasting, Feature Engineering.

Data & Modeling: Statistical Modeling, Experimentation, High-Dimensional Regression, Optimization, End-to-End Pipelines, Model Deployment.

Selected Projects

End-to-End Equity Modeling Pipeline: Architected a modular, spec-driven (**YAML**) **Python** pipeline for portfolio construction from universe of S&P 500 assets: pluggable covariance/return estimators via decorator registries, **Numba-JIT** Kalman filters, **CVXPY** optimization, a walk-forward backtest engine, and a CLI rebalancer that emits live trades. Built on clean ABCs with unit tests and reproducible configs; runs the full **S&P 500** (daily/weekly data since 2010) in ~5–10 min per strategy.

ML Surrogates for Derivative Pricing:  Trained **Gaussian Process** (Matérn) and **PyTorch** neural-network surrogates to emulate Monte Carlo and finite-difference (Crank-Nicolson) solvers for the **Heston** and **CEV** models, recovering prices and **Greeks** (via autograd) at a fraction of the numerical cost.

Portfolio Aggregator (production web app): Built and self-hosted a **Dockerized FastAPI** dashboard that aggregates multi-brokerage accounts through a third-party **REST API** (SnapTrade) into a SQLite store, with scheduled auto-sync, historical snapshots, and daily-change tracking.

Wellbore Geology Prediction (Kaggle): Built the geologic-surface stage of a two-stage pipeline predicting subsurface layer depth along horizontal wells: **SPDE Matérn** Gaussian-random-field surfaces fit over nearest calibration wells with a multi-marker ensemble, composed with an HMM-style **Viterbi** dynamic-programming refinement and Bayesian smoothness-averaging. Multi-marker surfaces substantially improved local out-of-fold RMSE.

Experience

University of California, Santa Barbara

Teaching Assistant & Graduate Student Mentor

Santa Barbara, CA

Sept. 2017 – June 2025

Taught statistics, probability, regression, time series, and stochastic processes; mentored undergraduate research projects in modeling and optimization, including posters and written reports.

Epic Systems

Technical Services Engineer

Madison, WI

Sept. 2014 – Sept. 2015

Partnered with large institutional clients to deploy and customize the Epic EHR in production, diagnosing issues and shipping codebase improvements under high-availability and regulatory constraints.

Prozess Technologie

Intern

St. Louis, MO

May 2013 – May 2014

Built computational simulations to validate laboratory equipment and calibrated pharmaceutical manufacturing hardware to industry standards.

Research & Publications

Banerjee, T., **Bernstein, A.**, Feinstein, Z. (2025). “Dynamic clearing and contagion in financial networks.” *European Journal of Operational Research* 321(2), 664–675.

Bernstein, A., Shkolnik, A. (2025). “Asymptotics of Quadratic Forms on a Simplex.” *In preparation*.